

1. Administrative & Study Identification

Full Study Title: AN INTERVENTIONAL, OPEN-LABEL, RANDOMIZED, MULTICENTER, PHASE 2 STUDY OF PF-07220060 PLUS LETROZOLE COMPARED TO LETROZOLE ALONE IN POSTMENOPAUSAL WOMEN 18 YEARS OR OLDER WITH HORMONE RECEPTOR-POSITIVE, HER2-NEGATIVE BREAST CANCER IN THE NEOADJUVANT SETTING **Brief/Lay Title:** A Study to Learn About the Study Medicine PF-07220060 Together With Letrozole Compared to Letrozole Alone in Women Post Menopause **Short Title/Acronym:** Neoadjuvant Phase 2 Study of PF-07220060 (Atirmociclib) + Letrozole in Breast Cancer **Protocol Number:** C4391025 **NCT Number:** NCT06465368 **Posting ID:** 717010061716090710 **Sponsor/Company Name:** Pfizer **Investigational Product:** PF-07220060 (Atirmociclib) and Letrozole **Phase of Development:** Phase 2 **Trial Status:** COMPLETED **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of preoperative atirmociclib (PF-07220060) plus letrozole versus letrozole alone on Ki-67 tumor expression, specifically measuring the rate of Complete Cell Cycle Arrest (CCCA) at Day 14.

Secondary Objectives: To evaluate the incidence of adverse events (AEs), circulating tumor DNA measurements, plasma concentrations of atirmociclib, and centrally assessed Ki-67 immunohistochemistry.

2.2 Background & Rationale To address the need for optimized targeted therapies in HR+/HER2- breast cancer. While dual CDK4/6 inhibitors are effective, they often cause dose-limiting hematologic toxicities (like neutropenia) largely driven by CDK6 inhibition. Atirmociclib is a highly selective CDK4 inhibitor designed to induce cell cycle arrest in breast cancer cells while sparing CDK6, potentially reducing hematologic toxicities and allowing for more continuous, effective dosing. This "window of opportunity" study assesses its preoperative efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Letrozole alone) **Blinding:** Open-label **Randomization:** 1:1 Randomized **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target \$N\$: 121 patients **Number of Sites:** >80 proposed sites across 16 countries (Global)

Estimated Study Start: July 2024 Estimated Primary Completion:
July 10, 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Postmenopausal women aged ≥ 18 years. 2. Histologically confirmed HR-positive and HER2-negative breast cancer (BC). 3. Baseline tumor Ki-67 score $\geq 10\%$ with unilateral, invasive T1c-T4c, N0-N2, M0 BC. 4. Treatment naïve for the treatment of BC; willing to undergo a baseline and Day 14 biopsy.

4.2 Exclusion Criteria 1. Prior systemic therapy, radiation, surgery, or investigational therapy for the treatment of breast cancer. 2. Certain severe medical conditions within the previous 6 months (e.g., myocardial infarction, severe unstable angina, symptomatic congestive heart failure). 3. Use of hormone replacement therapy (HRT) or estrogen-containing medication within 2 weeks prior to diagnostic tissue sample.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** PF-07220060 (Atirmociclib) 300 mg given as tablets by mouth twice a day plus Letrozole (Trade Name: Femara, ATC Code: L02BG04) 2.5 mg by mouth once a day. * **Active Comparator Arm:** Letrozole (Trade Name: Femara, ATC Code: L02BG04) 2.5 mg given as tablets by mouth once a day. **Route of Administration:** Oral **Duration of Treatment:** 14 days continuously.

5.2 Concomitant & Prohibited Therapy Permitted: Continuation onto standard-of-care therapy (e.g., endocrine therapy, surgery, radiotherapy, chemotherapy) per physician's choice after the 14-day study period. **Prohibited:** Prior systemic therapies and concurrent estrogen-containing medications during the trial window.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day -28 to 0 Informed Consent, Medical History, Baseline Biopsy Baseline Day 0 Randomization, First Dose Intervention/Interim Days 1 to 13 Safety Assessments, Blood Draws, Investigational Product administration End of Study Tx Day 14 Final Study Dose, On-treatment Biopsy (Centrally Assessed) Follow-up Day 42 (≈ 28 days post-dose) Safety Follow-up Visit, AE Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Rate of Complete Cell Cycle Arrest (CCCA), defined as Ki-67 $\leq 2.7\%$ at Day 14.

Measurement Method: Centrally assessed post-treatment tumor biopsy using Ki-67 immunohistochemistry.

7.2 Secondary & Exploratory Endpoints 1. Incidence of Adverse Events (AEs), serious AEs, and AEs leading to study drug discontinuation. 2. Circulating tumor DNA measurements at baseline and Day 14. 3. Ctrough and peri-biopsy plasma concentrations of atirrmociclib. 4. Spatial omics assessment of changes in the tumor microenvironment (Exploratory).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via clinical evaluations, blood draws, and patient interviews by site staff over the 14-day treatment and 28-day follow-up periods. **Serious**

Adverse Events (SAEs): Standard protocol tracking for all treatment-emergent SAEs leading to discontinuation. **Data**

Monitoring Committee (DMC): Independent Data Monitoring Committee (DMC) monitors safety, toxicity (e.g., neutropenia), and scientific integrity throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for primary endpoint evaluations (CCCA rates); Safety Analysis Set for AEs.

Sample Size Justification: Target enrollment of 121 randomized 1:1, appropriately powered to detect significant reductions in Ki-67 expression in the atirrmociclib + letrozole arm versus the letrozole-alone arm. **Handling of Missing Data:** Per standard Pfizer biostatistical guidelines for phase 2 oncology studies.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous oversight by clinical site monitors and the independent DMC. **Audit Trail:** Centralized tracking of Ki-67 assay scoring, electronic data capture (eCRF) locking, and standard clinical trial regulatory auditing.

11. Study Identifiers & Governance

Primary ID: C4391025 **Registry Number:** NCT06465368 **Posting ID:** 717010061716090710 **Sponsor/Lead Organization:** Pfizer **Study Title:**

Neoadjuvant Phase 2 Study of PF-07220060 (Atirmociclib) + Letrozole in Breast Cancer **Official Regulatory Title:** AN INTERVENTIONAL, OPEN-LABEL, RANDOMIZED, MULTICENTER, PHASE 2 STUDY OF PF-07220060 PLUS LETROZOLE COMPARED TO LETROZOLE ALONE IN POSTMENOPAUSAL WOMEN 18 YEARS OR OLDER WITH HORMONE RECEPTOR-POSITIVE, HER2-NEGATIVE BREAST CANCER IN THE NEOADJUVANT SETTING

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Hormone Receptor-Positive (HR+), HER2-Negative Breast Cancer **Diagnosis Threshold:** ER/PR-positive disease by IHC or ISH; Ki-67 score $\geq 10\%$ **Medical Methodology:** Neoadjuvant "Window of Opportunity" targeted endocrine therapy + CDK4 inhibition **Optimization Target:** Complete Cell Cycle Arrest (CCCA) indicated by Ki-67 $\leq 2.7\%$

13. Study Procedures & Methodology

Management Pathway: 14-day preoperative window of opportunity treatment sequence **Biopsy Protocol:** Required baseline diagnostic biopsy and matched Day 14 on-treatment biopsy **Evaluation Protocol:** Centralized pathology review for Ki-67 immunohistochemistry to avoid local variance **Quality Audit Mechanism:** Continuous monitoring of adverse hematologic toxicities (e.g., cytopenia/neutropenia tracking) and protocol adherence.

14. Observation & Follow-Up Schedule

Total Duration: Up to 70 days (28 days screening + 14 days treatment + 28 days follow-up) **Onsite Visit Cadence:** Baseline (Day 0), Treatment End/Biopsy (Day 14), Follow-up (Day 42) **Remote Monitoring Frequency:** Intermittent remote check-ins for AE monitoring based on investigator discretion. **Checklist Review Interval:** Daily during the 14-day active oral dosing period.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				